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Received 30 October 2023
Revised 11 December 2023
21 December 2023
23 December 2023
Accepted 20 January 2024

Building new business models in MSMEs: digital capabilities, global value chains integration and the moderating role of entrepreneurial failure

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Abstract

Purpose – The research aims to explore the relationships between the digital capabilities of micro, small and medium enterprises (MSMEs), their participation in global value chains and their adoption of innovative business models. Additionally, the study investigates how the prior experiences of entrepreneurial failure may influence or moderate these relationships.

Design/methodology/approach – To address these research objectives, the study draws upon data obtained from the Flash Eurobarometer 486 survey, a comprehensive dataset that explores the challenges faced by 13,197 European MSMEs as they navigate the complexities of growth, the integration of sustainable business models and the incorporation of digital technologies. To test the proposed hypotheses, the research employs multivariate logistic regression analysis.

Findings – Digital capabilities are found to be positively associated with business model innovation, while integration into global value chains is linked to a higher likelihood of implementing new business models. Interestingly, entrepreneurial failure did not significantly influence the relationship between digital capabilities and the adoption of new business models. In contrast, entrepreneurial failure significantly moderated the impact of global value chain inclusion on business model innovation, particularly in MSMEs with a history of failure.

Originality/value – This article provides practical guidance to entrepreneurs and companies interested in enhancing their digital strategies and engagement in global value chains, considering the entrepreneurs' business histories.

Keywords Business models innovation, Dynamic capabilities, Digital capabilities, Entrepreneurial failure, Global value chains, Micro, small and medium enterprises

Paper type Research paper

1. Introduction

In the dynamic context of business, micro, small and medium enterprises (MSMEs) emerge as essential pillars of economic growth and innovation (Singh *et al.*, 2022). These companies, known for their agility and adaptability, play a vital role in the development of global economies

Pedro Mota Veiga and Sandra Marnoto would like to thank to NECE – Research Centre for Business Sciences – funded by the Multiannual Funding Programme of R&D Centres of FCT – Fundação para a Ciência e a Tecnologia – under the project "UIDB/04630/2020".

Journal of Small Business and Enterprise Development
Vol. 31 No. 3, 2024
pp. 492-514
© Emerald Publishing Limited
1462-6004
DOI 10.1108/JSBED-10-2023-0501

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and in responding to the demands of the digital age (Santos *et al.*, 2023; Singh *et al.*, 2022). In a world where technology advances incessantly and markets globalize at a rapid pace, MSMEs face a landscape filled with promising opportunities and complex challenges that can define the success or failure of their trajectories (Varma and Dutta, 2023; Venkatesh *et al.*, 2021). In this ever-changing scenario, MSMEs not only drive innovation but also foster job creation, stimulate local development and significantly contribute to sustainable economic growth (Ferreira *et al.*, 2020). Their ability to respond quickly to market changes and consumer demands makes them essential players in today's globalized economy (Santos *et al.*, 2023). However, this adaptability also means that these businesses are continuously confronted with new challenges, from the effective integration of innovative technologies to fierce international competition (Varma and Dutta, 2023). Some challenges have an internal origin. Many MSMEs are family-owned businesses, and they have to deal with succession problems at some point. However, they develop better capabilities to use scarce resources efficiently to produce innovation, which is critical for business success and the effectiveness of succession processes (Baltazar *et al.*, 2023). Therefore, the ability of MSMEs to thrive and innovate not only influences their fate but also plays a crucial role in the economic development of entire regions. How these companies face and overcome the challenges of the digital age not only shapes their success but also contributes to the global economic landscape, influencing how businesses are conducted and innovations are adopted worldwide (Varma and Dutta, 2023; Venkatesh *et al.*, 2021).

Building new business models in MSMEs can be a crucial step in adapting to changing market conditions and achieving growth. Innovative business models for MSMEs cover a wide range of strategies, including customer-centricity, digital transformation, innovation and collaboration, sustainability and corporate social responsibility, optimization of the value chain, sharing economy and peer-to-peer models (Ferreira *et al.*., 2022; Huikkola *et al.*., 2022). Digital capabilities are a critical factor in the competitiveness, innovation and growth of MSMEs, enabling them to be better positioned to navigate the challenges and opportunities presented by the digital economy (Aloulou *et al.*, 2023; Balta *et al.*, 2023; Wang *et al.*., 2023). They are understood as the organization's ability to effectively leverage digital technologies and tools to enhance various aspects of their operations, encompassing a range of skills, resources and strategies (Balta *et al.*, 2023; Wang *et al.*., 2023; Eechoud and Ganzaroli, 2023). Global value chain integration opens new opportunities, diversification and learning experiences that can shape innovative approaches. Entrepreneurial failure, when managed effectively, serves as a valuable feedback mechanism for refining and strengthening new business models. These factors collectively contribute to the adaptability and competitiveness of MSMEs in an ever-changing business landscape (Gao *et al.*, 2022).

Based in the resource-based view (RBV) of the firm, global value chains (GVCs) serve as intricate networks where multinational enterprises from advanced economies collaborate with suppliers and partners in developing nations (Kano *et al.*, 2020; Siaw and Okorie, 2022; Siaw and Sarpong, 2021). Smaller firms embedded in GVCs thrive by actively participating in multiple networks, enhancing their learning, and development of capabilities, ultimately maximizing their competence in global markets and improving their value creation and appropriation within GVCs (Siaw and Okorie, 2022). This participation fosters adaptability, enabling them to navigate diverse relationships on technology-enabled platforms and allocate resources effectively within these connections. Adopting a network perspective, it becomes evident that a firm's business model is shaped by strategic choices regarding the creation and capture of value within the network. A firm's role within this network significantly influences its unique business model (Shafer *et al.*, 2005). Consequently, the involvement of MSMEs within GVCs significantly enhances their inclination to embrace new business models, particularly in the context of the rapid technological advancements brought about by Industry 4.0 (Siaw and Okorie, 2022; Venkatesh *et al.*, 2021).

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Entrepreneurial pathways often encompass a sequence of attempts, failures and subsequent efforts, ultimately leading to success (Eklund *et al.*, 2020). Entrepreneurial failure is a multifaceted phenomenon with various definitions and interpretations in the literature (Czakon *et al.*, 2022). Failure is often associated with prolonged organizational decline and financial deterioration (Ayamga *et al.*, 2024). However, it frequently occurs in the early stages of new ventures when the organizational structure collapses (Petrucci and Milanesi, 2022). Successful entrepreneurs often view failure as a significant and instructive event in their lives, enriching their experiences and enhancing future endeavors (Petrucci and Milanesi, 2022). Petrucci and Milanesi (2022) emphasize that many successful ventures emerge from previous projects or enterprises that experienced failure, underscoring the critical role of the post-failure phase in the entrepreneurial process. The post-failure phase represents a pivotal stage in the entrepreneurial journey, characterized by profound changes, new opportunities and fresh starts (Petrucci and Milanesi, 2022). Examining entrepreneurial failure in the context of a business network, such as GVCs, involves analyzing its impact on entrepreneurs' decisions. When a venture fails, it doesn't simply dissipate; instead, it releases residual resources – what remains after the failure – which can flow through the network toward new setups, projects or ventures. According to Aaboen *et al.* (2011), commencing and nurturing new relationships require substantial investments of effort and resources over time. This entails exploring diverse networks, identifying and approaching potential partners and effectively managing the intense flow of social-material interactions. The networking capability of a new venture appears to emerge more from learning through failures than from successful interactions (McGrath *et al.*, 2019). This learning process unfolds through active engagement, serving as a core capability that shapes the strategic networking and positioning of the new venture. The high failure rate of initial relationships implies that the networking capability of a new venture is shaped by intricate and enduring learning dynamics (McGrath *et al.*, 2019). The volatile nature of initial relationships is often overlooked, and the role of the entrepreneur after the failure of a new venture is underestimated. McGrath *et al.* (2019) emphasize that in this process, the new venture can progressively learn the capability to manage and use network relationships to its advantage until overcoming the initial managerial lack of knowledge and experience.

The main objective of our research is to examine how digital capabilities and integration into global value chains impact the adoption of new business models by MSMEs, exploring the moderating role of entrepreneurial failure experiences. To address these objectives, we utilized data from the Flash Eurobarometer 486 survey titled "SMEs, Start-ups, Scale-ups, and Entrepreneurship." The Eurobarometer survey collected responses from 13,197 MSMEs through telephone interviews conducted between February and May 2020. The research employed a multivariate logistic regression analysis to test the proposed hypotheses.

Our research can offer several valuable contributions to the understanding of MSMEs and their role in the evolving business landscape. Firstly, the study can clarify the pivotal role of MSMEs in economic growth and innovation, emphasizing their adaptability and significance in responding to the demands of the digital age. By exploring the challenges and opportunities faced by MSMEs in the dynamic global market, the research can contribute with knowledge of how these enterprises drive innovation, create employment and foster sustainable economic growth.

The article can also contribute to the discourse on innovative business models for MSMEs, providing a comprehensive overview of strategies such as customer-centricity, digital transformation, sustainability and global value chain integration. It highlights the critical role of digital capabilities in enhancing the competitiveness, innovation and growth of MSMEs, positioning them to navigate the challenges of the digital economy effectively. Building on the RBV of the firm and the concept of GVCs, the research can offer a deeper understanding of how MSMEs embedded in GVCs thrive through active participation in

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diverse networks. This involvement enhances their learning, development of capabilities and competence in global markets. The study's network perspective emphasizes how a firm's business model is shaped by strategic choices within the network, particularly in the context of Industry 4.0. Examining entrepreneurial failure within the context of GVCs, the research can offer valuable outlooks into the impact of failure on entrepreneurs' decisions and how residual resources from failed ventures can catalyze innovation and growth when combined with digital capabilities.

In the sections that follow, we will outline the framework of this research. Section 2 will encompass a review of relevant theories, offering insights into the existing body of knowledge on the subject, followed by the development of hypotheses (Section 3). In Section 4 we provide a comprehensive overview of the methodology employed in our study, the research design, data collection and analysis techniques. In Section 5, we expose the findings of our research, presenting empirical evidence and statistical outcomes, which serve as the cornerstone for our subsequent discussion. In Section 5, the concluding part, we dedicated to the discussion of our results, addressing both their theoretical and practical implications. We will also acknowledge the limitations of our study and discuss potential future research directions.

2. Theoretical underpinnings

2.1 Business models innovation

The definition of a business model is not straightforward. Zott *et al.* (2011) made a comprehensive literature review about the concept and concluded that it has been applied in different phenomena, i.e., in diverse questions, contexts and management domains, such as firms' value creation and capture, various types of e-business and workings of technology innovation. In this work, we assert that a business model may be described as a framework that defines how value is created, delivered and captured by a firm. At the center of the model is the value proposition of the business, i.e., the distinctive combination of products and services that a firm offers to its customers. It expresses how the products and services provided solve specific problems or fulfill the needs and desires of the customers. It clarifies why the solutions of a firm are valuable to its customers and what sets them apart from competitors (Osterwalder and Pigneur, 2010). A business model innovation exists when a firm designs a new business model that has at least one feature that is considerably different from others. This feature may refer, for example, to the supply of a new product and service, the provision of existing products and services using a new approach, and the creation of a new industrial field. It is recognized as an important form of fostering the development and competitiveness of a firm (May and Zang, 2021). The 21st century has been fertile in business model innovations. Many researchers describe how industry 4.0 (i4.0) technologies, such as additive manufacturing, agile robotics, artificial intelligence, biotechnology, blockchain, the Internet of things, quantum computing and nanotechnology, create radically new entrepreneurial opportunities that demand somewhat disruptive new business models, transforming products, services and global markets (Rayna and Striukova, 2014). Added manufacturing, for instance, has the potential to radically change the mass production paradigm into a mass customization age (Rosenbaum *et al.*, 2021). The use of 3D printing on a wide, large scale might not only change the way production is organized but also where it takes place as it might become, once again, more localized. Hence, if this turns out to be a reality, as it seems, the present strategic business models will need to be revised, as well as the current approaches to international manufacturing and marketing (Rosenbaum *et al.*, 2021; Hannibal, 2020).

2.2 Digital capabilities

According to Teece (2007), a company's dynamic capabilities are those that enable the integration, construction and reconfiguration of internal and external resources/competencies

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to deal with and potentially shape ever-changing business environments. The dynamic capabilities view (Teece *et al.*, 1997) explains how companies respond to rapid and technological changes, extending the RBV (Barney, 1991, 2001) that helps understand how different, difficult-to-imitate resources are combined to create capabilities that generate superior performance (Balta *et al.*, 2023). Teece (2016) argues that a dynamic capability is a higher-order capability that serves as a theory of competitive advantage for a company operating in a highly dynamic and turbulent environment. Organizations often develop dynamic capabilities by integrating, building and reconfiguring internal and external competencies when operating in highly competitive and dynamic environments (Dejardin *et al.*, 2022). The integration of digital technology has transformed traditional dynamic capabilities, creating multiple dimensions of digital capabilities with an impact on entrepreneurship and sustainable business model innovation (Anwar *et al.*, 2022; Evans *et al.*, 2017; Gao *et al.*, 2022; Wang *et al.*, 2023). The practical application of digital technology to dynamic capabilities, particularly in stages ranging from creativity to production, plays a crucial role in data collection and processing, formulating new value propositions and promoting an efficient and cyclical value creation process (Audretsch *et al.*, 2022; Balta *et al.*, 2023; Gao *et al.*, 2022).

2.3 Global value chains

The importance of interorganizational relationships (Richardson, 1972) and more specifically of networks of interfirm relationships (Ford *et al.*, 2011), such as a GVC, can be traced back and explained by the RBV of the firm (Barney, 1991, 2001; O'Farrell and Hitchens, 1988; Penrose, 2009), according to which no firm has on its own all the resources and capabilities it needs to reach its objectives. Quite the opposite, the resources that allow a firm to gain and sustain a competitive advantage are idiosyncratic, valuable, rare and difficult to imitate or substitute (Barney, 1991, 2001). Consequently, firms access the resources and capabilities for which they do not have a competitive advantage through their partners in the relationships they create, develop and maintain in their business networks (Ford *et al.*, 2011; Richardson, 1972). Contrary to traditional approaches based on the transaction costs theory, productive knowledge is not evenly spread in an industry, nor the costs of production are the same everywhere Loasby and Foss, 1998). This is exactly the main reasoning underlying GVC (De Marchi *et al.*, 2020). GVC are international interorganizational networks through which multinational businesses located in advanced economies engage in relationships with suppliers and other partner firms in developing and emerging countries. The objective of those multinationals is to minimize the production costs of products that are manufactured by their partners in the GVC. These products will later be sold for maximum profits in advanced economies (Kano *et al.*, 2020; Siaw and Okorie, 2022; Siaw and Sarpong, 2021).

2.4 Entrepreneurial failure

Entrepreneurial failure is an important and complex phenomenon (Khelil, 2016). Most probably because of its complexity, there is a multitude of definitions and conceptualizations in the literature, leaving us yet to find a clear concept. We will follow Czakon *et al.* (2022) and define entrepreneurial failure as a situation when an individual entrepreneur involuntarily terminates his or her relationship with an entrepreneurial venture because it is no longer possible to achieve an appropriate entrepreneurial performance. Often failure is perceived as the result of years of organizational decline and deterioration of financial performance (Ayamga *et al.*, 2024). However, most new ventures tend to fail during the early stage of their development (Petrucci and Milanesi, 2022). Here we refer to 'new venture failure" as a situation when the organizational structure of a new venture collapses, therefore terminating its activities and relationships with its counterparts (Petrucci and Milanesi, 2022). Research shows that 'new venture failure" is largely due to market changes and risk-taking (Mai *et al.*, 2021). However, as Eklund *et al.* (2020) put it,

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this is not surprising, when taking into consideration two elements in which most definitions of 'entrepreneur' are rooted: the discovery of new successful opportunities or combinations (Schumpeter, 1934) and the management of uncertainty (Knight, 2002).

3. Hypothesis development

3.1 Digital capabilities and business models innovation

The incorporation of information technology changed the concept of the business model from a traditional concept to a digital or innovative one (Kajtazi *et al.*, 2023). An example of how business models changed in the last years, due to recent technological advances, is the integration of I4.0 technologies and the principles of the circular economy (Agrawal *et al.*, 2022). These new business models aim not only to create economic value but also social and environmental values (Gregori and Holzmann, 2020). The I4.0 technologies have also given rise to new business models in the sharing economy through the development of technology-enabled platforms that end up serving as both an intermediary for sharing resources, such as Uber and Airbnb, and a marketplace, like Amazon (Mai *et al.*, 2021).

Digital capabilities in MSMEs play a central and transformative role in business models innovation, once they enable organizations to create new ways of delivering value, reaching customers and differentiating themselves in the marketplace (Gao *et al.*, 2022). They empower organizations to be more agile, customer-focused and responsive to the opportunities and challenges of the digital age (Wang *et al.*., 2023). According to Balta *et al.* (2023), "digital capability refers to the ability of a firm to use digital technologies to achieve a competitive advantage." In a rapidly changing business environment, businesses that leverage digital capabilities effectively are more likely to innovate and adapt their business models to meet evolving customer needs and market dynamics. Digital capabilities can be divided into different types: digital sensing, digital seizing and digital transformation (Balta *et al.*, 2023; Gao *et al.*, 2022; Huikkola *et al.*., 2022). The emergence and development of these capabilities in an organization is activated by the conditions of the external and internal environment and the available digital technologies and conditioned by the organization's digital culture (Balta *et al.*, 2023; Proksch *et al.*, 2021). Such capabilities enable companies to continuously gather intelligence, develop new value propositions and create innovative business models. They are also instrumental in the rise of the innovation and adventure spirit, promoting entrepreneurship, which encourages the design of innovative business models (Gao *et al.*, 2022). In turn, companies that invest in the development and strengthening of their digital capabilities (basic, operational and integration) perform better, which suggests that they are in a better position to develop more efficient and innovative business models (Wang *et al.*., 2023). The development of digital capabilities was also identified by Anwar *et al.* (2022), as an indirect driver for internationalization, through innovation in business models, that support the entrance of MSMEs manufacturers in global and more profitable markets. Given the current concerns of governments and consumers, the transition from linear to circular economy is a battle that MSMEs will have to face shortly, which necessarily requires the definition of new business models that can offer alternatives to close the biological and technological cycles that provide the groundwork for this shift. Digital technologies and digital capabilities are critical to this process by fostering innovation is circular business models (Eechoud and Ganzaroli, 2023).

The interplay between strategic and dynamic capabilities, with the exploration of digital capabilities, helps companies modify their strategic capabilities in the transition to smart (Bresciani *et al.*, 2021; Huikkola *et al.*, 2022). Digital capabilities play a critical role in promoting innovation in business models, which, in turn, contributes to more robust business performance (Wang *et al.*, 2023). As digitalization enables the creation of various business models for technology companies, firms reconfigure their strategic capabilities and processes

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when becoming providers of smart solutions (Huikkola *et al.*, 2022; Wang *et al.*, 2023). The transition to smart solutions offers economic and strategic benefits for companies, including cross-selling opportunities, cost savings and improved customer experiences (Balta *et al.*, 2023; Huikkola *et al.*, 2022).

Business model innovation acts as an intermediary in the relationship between digital capabilities and business performance (Bresciani *et al.*, 2021; Wang *et al.*, 2023). Thus, the ability to innovate in business models amplifies the positive impact of digital capabilities on company performance (Anwar *et al.*, 2022; Wang *et al.*, 2023). Companies with strong digital capabilities are more likely to develop innovative and effective business models. It is important to note that while there is a general trend that digital capabilities drive business model innovation and enhance business performance, a significantly small portion of companies can fully leverage these digital capabilities to improve their business value (Bresciani *et al.*, 2021; Wang *et al.*, 2023).

The complexity of the business model innovation process is acknowledged, emphasizing the need to understand the factors that influence this digital transformation. From this analysis, we formulate the first hypothesis of this research as follows:

H1. The digital capabilities of MSMEs have an impact on the adoption of new business models

3.2 Global value chains and business models innovation

Multinational enterprises are the dominant firms in the GVC and, as such, they coordinate the creation and capture of value in the network (Kano *et al.*, 2020; Siaw and Okorie, 2022; Siaw and Sarpong, 2021). They exert power over their counterparts by setting products and services as the basis of value offerings and, thus, enjoying lower production costs. Yet, the emergence of the sharing economy is changing the basis of value offerings in GVCs (De Marchi *et al.*, 2020). Digital technological advancements are moving the basis of value offerings from products and services to platforms (Nambisan *et al.*, 2019). Those platforms ease the existence of interactions in multi-actor networks. The multinationals' partners can, therefore, participate in other networks, hence becoming simultaneously their partners and their competitors (Siaw and Okorie, 2022). In conclusion, the advent of the sharing economy and technology-enabled platforms is shifting the source of value co-creation from controlling key resources to controlling platforms that allow access to key resources from and by firms in geographically dispersed locations (Siaw and Okorie, 2022).

In the face of this shift, multinational enterprises must develop both operational efficiencies and strategic effectiveness. First, multinationals must invest in the flexibility for cooperating with their partners while differentiating from them. These operational efficiencies stress the economic and transactional aspects related to geographical mapping. Second, multinationals must develop effective strategies to optimize value creation and capture by focusing on the exploitation of micro-specialized skills and capabilities by their partners (Kano *et al.*, 2020; Siaw and Okorie, 2022; Siaw and Sarpong, 2021). On the other side of the relationships, research shows that the smaller firms in a GVC benefit from engaging in multiple networks to better learn and develop their capabilities, and maximize their competencies on global markets, thus improving value creation and appropriation in GVC (Ivarsson and Alvstam, 2010; Schmitz and Knorringa, 2000). This active participation in several networks and platforms increases the responsiveness of business models by firms, because of both the firm's ability to manage different relationships on technology-enabled platforms and to allocate resources effectively in those relationships (Siaw and Okorie, 2022).

Assuming a network perspective, the business model is the result of the strategic choices made by the firm regarding the creation and capture of value in a network. The creation and capture of value are a result of the interactions of heterogeneous firms that act and react

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within a business network. The role that a firm plays within the network is a key component of its business model (Shafer *et al.*, 2005). Therefore, the participation of micro, small and medium firms in GVC makes them more prone to adopt new business models, particularly in the face of the new technological developments of the i4.0 and the incentive to participate in multiple platforms. From this analysis, we formulate the second hypothesis of this research as follows:

H2. Inclusion in global value chains by MSMEs has an impact on the adoption of new business models

3.3 Entrepreneurial failures, digital capabilities and business models innovation

Entrepreneurial failure is an important and complex phenomenon (Khelil, 2016). Most probably because of its complexity, there is a multitude of definitions and conceptualizations in the literature, leaving us yet to find a clear concept. We will follow Czakon *et al.* (2022) and define entrepreneurial failure as a situation when an individual entrepreneur involuntarily terminates his or her relationship with an entrepreneurial venture because it is no longer possible to achieve an appropriate entrepreneurial performance. Often failure is perceived as the result of years of organizational decline and deterioration of financial performance (Ayamga *et al.*, 2024). However, most new ventures tend to fail during the early stage of their development (Petrucci and Milanesi, 2022). Here we refer to 'new venture failure" as a situation when the organizational structure of a new venture collapses, therefore terminating its activities and relationships with its counterparts (Petrucci and Milanesi, 2022). Research shows that 'new venture failure" is largely due to market changes and risk-taking (Mai *et al.*, 2021). However, as Eklund *et al.* (2020) put it, this is not surprising, when taking into consideration two elements in which most definitions of 'entrepreneur" are rooted: the discovery of new successful opportunities or combinations (Schumpeter, 1934) and the management of uncertainty (Knight, 2002). Numerous successful entrepreneurs had previously gone through several trials and failures before thriving (Eklund *et al.*, 2020). Furthermore, successful entrepreneurs tend to depict failure as a dramatic, but valuable event in their lives, as it provides an enriching experience that teaches them how to enhance later entrepreneurial activities (Petrucci and Milanesi, 2022). In short, most entrepreneurs have a difficult path but the dynamic of failing and starting over might be positive, as it provides not only learning but also a different recombination of resources until a more feasible combination is found. Petrucci and Milanesi (2022) conclude that most of the new successful ventures rise from previously failed projects or ventures and argue that the post-failure phase is a critical stage in entrepreneurial processes. Dana *et al.* (2021) emphasized the need to analyze and assess the causes of failures in innovation processes, at the risk of missing opportunities for corrective learning and reinforcing the leading role of success narratives, sometimes due to favorable conditions.

Instead of viewing past entrepreneurial experience as a personal failure, resilient entrepreneurs with negative entrepreneurial experiences see it as an opportunity for learning and the development of the necessary resilience to operate in increasingly complex and challenging business environments (Lafuente *et al.*, 2019). Even if past entrepreneurial experiences were negative, they can have a significant impact on subsequent international orientation and business model innovation (Audretsch *et al.*, 2022; Jenkins *et al.*, 2014; Proksch *et al.*, 2021). Resilient entrepreneurs can capitalize on the accumulated knowledge from their past entrepreneurial experiences (Coelho and McClure, 2005), raising questions about the role of generative learning in the contribution of resilient entrepreneurs to their subsequent ventures (Corner *et al.*, 2017; Khelil, 2016).

Entrepreneurial failure, when coupled with digital capabilities, can lead to business model innovation by fostering a culture of learning, adaptability, and customer-centricity. Digital

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tools and data-driven insights provide the means to turn setbacks into opportunities for improvement and business model innovation. The period during and after COVID-19 has given rise to renewed interest in understanding the relationship between the digitalization of companies' activities and business failure. The use of digital technologies is key to increasing the resilience of MSMEs during periods of crisis, namely by using digital capabilities to enable companies to explore new business opportunities beyond their traditional boundaries (Khurana *et al.*, 2022). Capitalizing on digital capabilities has helped prevent business failures during the pandemic and allowed companies to respond to external shocks and induce new business strategies (Zia *et al.*., 2023). However, the digitalization of companies' processes, products and services also adds volatility to their activity and interactions with stakeholders. By influencing business models, digital technologies create opportunities and respond to challenges. At the same time, they are disruptive to operational processes, partnerships and old technologies, which can bring risks of failure (Khurana *et al.*, 2022). MSMEs with constrained resources may not be able to migrate quickly from old technologies to new ones, not least because they are limited in the development of their digital capabilities. Digital transformation, from micro to macro levels, is also influenced by the characteristics of entrepreneurial ecosystems (Guimarães *et al.*, 2023).

The relationship between entrepreneurial failure, digital capabilities and business model innovation is complex and intertwined. Entrepreneurial failure can catalyze innovation and growth when combined with digital capabilities. However, the literature shows that while digital capabilities are important for increasing the resilience of organizations to changes in their environment, it is not clear whether this role is affected by the entrepreneur's prior experience, leading us to hypothesize the following:

H3. The entrepreneur's previous failure moderates the impact of the digital capabilities of MSMEs on the adoption of new business models

3.4 Entrepreneurial failures, global value chains and business models innovation

Attempting to deal with the phenomenon of entrepreneurial failure from the perspective of a business network, such as GVCs, means to understand what happens to the marginal resources and relationships that survive in the aftermath of the failure. The post-failure phase might be a critical step of the entrepreneurial process as profound change, opportunities and new beginnings can arise. Petrucci and Milanesi (2022) argue that when a venture fails, "it does not simply dissolve into a 'vacuum' rather it releases a set of 'residual resources' – namely, what remains after failure – which can 'flow' through the network towards new settings, projects or ventures." However, it must be noted that the new ventures that may follow can deeply diverge from past ventures, leading the entrepreneur to largely different activities or projects (Deichmann and Ende, 2014). Petrucci and Milanesi (2022) argue that the post-failure phase is "nonlinear" and full of possibilities. New unexpected interactions can appear because of the entrepreneur's motivation to use the residual resources and to restart a new enterprise, as well as its former partners' will to absorb marginal resources in the network. This process might provide a continuity of the past, but this continuity might be characterized by different relational hierarchies. Aaboen *et al.* (2011) propose that the initiation and cultivation of new relationships necessitate substantial investments in time, effort and resources. This entails exploring diverse networks, identifying and approaching potential partners and effectively managing the complex interplay of social-material interactions. Contrary to conventional perspectives, the networking capability of a new venture appears to stem more from learning through failures than from successful interactions (McGrath *et al.*, 2019). This process of learning unfolds actively, acting as a foundational capability that molds the strategic networking and positioning of the new venture. The considerable failure rate of initial relationships indicates that the networking

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capability of a new venture is shaped by intricate and enduring learning dynamics (McGrath *et al.*, 2019). The dynamic and often unpredictable nature of initial relationships is frequently overlooked, and the post-failure role of the entrepreneur in a new venture is commonly undervalued. McGrath *et al.* (2019) underscore that within this process, the new venture progressively acquires the ability to effectively manage and leverage network relationships, surpassing the initial lack of managerial knowledge and experience.

So, we might consider that this stage is a continuous of unexpected events where the entrepreneur needs to improvise and react to recover from failure and grab new opportunities. As such the entrepreneur might diverge from the past and follow a different path, avoiding previous mistakes and therefore being less prone to adopt new business models.

Hence, we formulate the fourth hypothesis of this research as:

H4. The entrepreneur's previous failure moderates the impact of inclusion in global value chains of MSMEs on the adoption of new business models.

In Figure 1, we present the conceptual model of research, which serves as a foundational framework for the entire study.

4. Methodology

4.1 Sample and data

To empirically test our hypotheses, we used data from the Flash Eurobarometer 486 survey "SMEs, Start-ups, Scale-ups and Entrepreneurship" (European Commission, 2020). These data focus on the obstacles and challenges that MSMEs face in Europe as they seek to grow, adopt more sustainable business models and integrate digitalization. This Eurobarometer collected responses from 13,197 MSMEs through telephone interviews conducted between 19 February and 5 May 2020. The research covered the 27 countries of the European Union and a further 8 European countries outside the EU, covering a wide variety of sectors. Academic studies have consistently used Flash Eurobarometer research due to its proven reliability and extensive validation over time (e.g. Ardito, 2023; Arranz *et al.*, 2023; Suchek *et al.*, 2023).

Regarding the distribution of companies (Table 1), 58.1% were micro companies (with one to nine employees), 26.6% were small companies (with ten to 49 employees) and 15.3% were medium-sized companies (with 50–249 employees). As far as the distribution by sector is

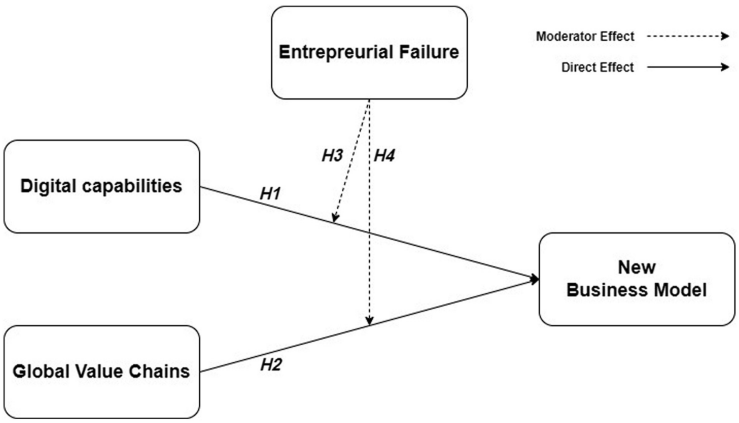

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Figure 1.
Conceptual model

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		N	%
Sector of activity (NACE) – sections	B – Mining and quarrying	42	0.3
	C – Manufacturing	2,598	18.6
	D –Electricity; gas; steam and air conditioning supply	67	0.5
	E – Water supply; sewerage; waste management/ remediation active	130	0.9
	F – Construction	1,380	9.9
	G – Wholesale and retail trade; repair of motor vehicles	3,975	28.5
	H – Transportation and storage	796	5.7
	I – Accommodation and food service activities	817	5.9
	J – Information and communication	552	4.0
	K – Financial and insurance activities	269	1.9
	L – Real estate activities	332	2.4
	M – Professional; scientific and technical activities	1,349	9.7
	N – Administrative and support service activities	606	4.3
	P – Education	355	2.5
	Q – Human health and social work activities	466	3.3
	R – Arts, entertainment and recreation	212	1.5
Employees	1 to 9 employees	8,108	58.1
	10 to 49 employees	3,709	26.6
	50 to 249 employees	2,129	15.3

Source(s): Authors' own creation

Table 1.
Sample characteristics

concerned, the companies were active in 16 different sectors, with the manufacturing sector (18.6%), retail companies (28.5%) and scientific and technical activities (9.7%) standing out. Table A1 in the Appendix shows the breakdown by country.

4.2 Measures

4.2.1 Dependent variable. The dependent variable is related to the adoption of a new business model (NBM). To measure this variable, we used a specific question in the questionnaire which asked: "In the last 12 months, has your company implemented a new management structure or adopted a different business model?" The answer to this question was binary, i.e. respondents could choose either "No" or "Yes".

4.2.2 Independent variables. To assess the company's digital capabilities, we used a specific question in the questionnaire: "Which of the following digital technologies has your company adopted?" The answers to this question were binary, i.e. respondents could choose between "No" or "Yes". The question contains several options: (1) AI, e.g. machine learning or technologies that identify objects or people, etc.; (2) cloud computing, i.e. storing and processing files or data on remote servers hosted on the Internet; (3) robotics, i.e. robots used to automate processes, e.g. in construction or design, etc.; (4) smart devices, e.g. smart sensors, smart thermostats, etc.; (5) big data analysis, e.g. data mining and predictive analysis; (6) high-speed infrastructure; and (7) blockchain. As suggested by Arranz *et al.* (2023), the digital capabilities variable (DIGITAL) was composed as a cumulative index of these seven categories of digital technologies (AI, cloud computing, robotics, smart devices, big data analytics, high-speed infrastructure and blockchain).

As for analyzing integration into global value chains, we adopted a measurement approach that incorporated the specific question from the questionnaire: "Is your company part of a global value chain?" The answer to this question was binary, allowing participants to choose between "No" and "Yes". The central aim of this question was to assess the level of participation of companies in these GVCs.

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4.2.3 Moderating variable. The moderating variable, related to previous episodes of failure on the part of the entrepreneur (FAIL), was assessed through the questions: "Have you ever closed – without bankruptcy – another company or companies of which you were the owner or co-owner?" and "Have you ever closed – due to bankruptcy – another company of which you were the owner or co-owner?". The answers to both questions were binary, allowing participants to choose between "No" and "Yes". The existence of previous failure on the part of the entrepreneur was considered if at least one of these situations had occurred.

4.2.4 Control variables. Several control variables were included to improve the predictive power of the modeling. These variables include the age of the company (AGE) (transformed using the Neperian logarithm); the size of the company in terms of the number of employees (EMPL) and turnover (TURNOVER) (both variables transformed using the Neperian logarithm); a set of dummy variables to control for the industrial sector; and a dummy variable equal to one if the company is based in a country belonging to the European Union.

Table 2 displays the variables incorporated into the research, offering a comprehensive overview of the key elements under analysis.

Measures	Variable	Question/Description	Type	Authors
Dependent variable	Adoption of NBM (Y/N)	In the last 12 months, has your company implemented a new management structure or adopted a different business model?	Binary	Arranz *et al.* (2023)
Independent variables	Digital capabilities	Which of the following digital technologies has your company adopted? (AI, cloud computing, robotics, smart devices, big data analytics, high	Composite (0–7)	Anwar *et al.* (2022)
	Integration into GVC	Is your company part of a global value chain?	Binary	Suchek *et al.* (2023)
Moderating variable	Entrepreneurial failure	Have you ever closed – without bankruptcy – another company or companies of which you were the owner or co-owner? Have you ever closed – due to bankruptcy – another company of which you were the owner or co-owner?	Binary	Hoogendoorn *et al.* (2019)
Control variables	Age of the company	Neperian logarithm transformation of the company's age	Continuous	
	Size of the company	Neperian logarithm transformation of the number of employees	Continuous	
	Turnover of the company	Neperian logarithm transformation of the turnover	Continuous	
	Industrial sector dummies	Dummy variables to control for the industrial sector	Dummy (Categorical)	
	EU	Dummy variables to control for firms located in the European Union countries	Dummy (Categorical)	

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Table 2. Variables and measures

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4.2.5 Econometric models. To evaluate our hypotheses, we chose to use multivariate logistic regression analysis. During this process, we developed two different models, both with the adoption of a new business model as the dependent variable. In the first model (Model 1), we used control variables as independent variables, along with digital capabilities and integration into global value chains. In the second model (Model 2), in addition to including the variables from the first model, we inserted variables that explore the interaction between digital capabilities and the entrepreneur's previous experience of failure, and between integration into global value chains and the entrepreneur's previous experiences of failure. These interaction variables were calculated by multiplying the values of the relevant variables. The function used in the logistic regression to estimate the probability of a given realization j (j = 1, . . .,n) of the dependent variable occurring, i.e. companies abandoning innovation activities, ($P[Y_j = 1] = \widehat{\pi}_j$), can be expressed by: $\widehat{\pi} = \frac{e^{X\beta}}{1+e^{X\beta}}$, where $\widehat{\pi}$ is the vector of estimated probabilities, X is the matrix of independent variables and β is the vector of logistic regression coefficients (Cameron and Trivedi, 2005). As the logistic model was used, the *Odds Ratio* (OR) was estimated. Linearizing this function with the *logit* transformation of the dependent variable gives the logistic regression econometric model under analysis, and the following model was estimated:

1. $Logit(\widehat{NBM}_j) = \beta_0 + \beta_1 CONTROL_j + \beta_2 DIGITAL_j + \beta_3 GVC_J$
2. $Logit(\widehat{NBM}_j) = \beta_0 + \beta_1 CONTROL_j + \beta_2 DIGITAL_j + \beta_3 GVC_J + \beta_4 FAIL + \beta_5 DIGITAL_j \times FAIL + \beta_6 GVC_J \times FAIL$

Because the data was collected in several different countries, violating the assumption of independence of the observations within each country, we used *clustered robust standard errors in* which the country corresponds to the *clustering* variable. We therefore did not include dummy variables associated with the company's country as a control variable. Secondly, we checked for the possible existence of multicollinearities between the variables used in the analysis, namely by analyzing the correlations between the variables used in the regression analysis and the value of the variance inflation factor (VIF). All the VIF values were less than 5.

5. Results

Table 3 contains the results of the descriptive statistics and the correlation between the variables used in the study. 15% of the MSMEs included in the sample had implemented new business models, 9% were included in global value chains and 10% of company managers had some previous experience of business failure.

The estimated logistic regression models (Model 1 and Model 2) that assess the hypotheses developed in this study are shown in Table 4. Both models have a reasonable predictive capacity (Model 1: AUC = 67.8%; Model 2: AUC = 68.1%), statistically significant

	Mean ± SD	Range	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) NBM	0.15 ± 0.36	0/1							
(2) AGM	2.91 ± 0.79	0/6.93	−0.004						
(3) EMPL	2.14 ± 1.38	0/5.52	0.122	0.253					
(4) TURNOVER	22.07 ± 10.48	0/34.54	−0.047	0.014	0.038				
(5) DIGITAL	1.34 ± 1.37	0/7	0.197	0.046	0.248	−0.089			
(6) GVC	0.09 ± 0.28	0/1	0.097	0.027	0.140	−0.055	0.201		
(7) FAIL	0.10 ± 0.3	0/1	0.027	−0.094	−0.108	−0.092	0.049	0.037	

Note(s): SD – Standard Deviation; Range – Minimum/Maximum
Source(s): Authors' own creation

Table 3. Descriptive statistics and matrix correlation

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Variables	Model 1 OR	CI 95%	p	Model 2 OR	CI 95%	p
Constant	0.12	[0.07–0.20]	0.000	0.11	[0.07–0.19]	0.000
AGE	0.88	[0.80–0.97]	0.010*	0.89	[0.80–0.99]	0.029*
EMPL	1.20	[1.14–1.26]	0.000**	1.20	[1.15–1.26]	0.000**
TURNOVER	0.99	[0.98–1.00]	0.004**	0.99	[0.98–1.00]	0.007**
EU	0.97	[0.66–1.43]	0.883	0.96	[0.65–1.43]	0.854
B – Mining and quarrying	0.38	[0.09–1.59]	0.186	0.38	[0.09–1.57]	0.181
D – Electricity; gas; steam and air conditioning supply	0.58	[0.27–1.24]	0.163	0.61	[0.28–1.32]	0.209
E – Water supply; sewerage; waste management/remediation activities	0.97	[0.62–1.52]	0.910	0.99	[0.63–1.54]	0.957
F – Construction	0.94	[0.77–1.15]	0.555	0.96	[0.79–1.17]	0.706
G – Wholesale and retail trade; repair of motor vehicles and	1.07	[0.91–1.26]	0.434	1.06	[0.90–1.25]	0.484
H – Transportation and storage	0.92	[0.70–1.22]	0.567	0.93	[0.70–1.24]	0.623
I – Accommodation and food service activities	1.41	[1.16–1.73]	0.001**	1.46	[1.21–1.77]	0.000**
J – Information and communication	1.19	[0.93–1.52]	0.171	1.19	[0.93–1.53]	0.175
K – Financial and insurance activities	1.28	[0.84–1.94]	0.249	1.29	[0.84–1.99]	0.239
L – Real estate activities	1.03	[0.73–1.45]	0.871	1.03	[0.73–1.46]	0.868
M – Professional; scientific and technical activities	1.08	[0.91–1.28]	0.392	1.10	[0.92–1.31]	0.282
N – Administrative and support service activities	1.17	[0.95–1.44]	0.138	1.08	[0.86–1.35]	0.491
P – Education	1.39	[0.96–2.00]	0.080	1.44	[0.98–2.11]	0.065
Q – Human health and social work activities	1.53	[1.17–2.00]	0.002**	1.53	[1.16–2.00]	0.002**
Arts, entertainment and recreation	1.28	[0.78–2.12]	0.333	1.30	[0.79–2.16]	0.302
DIGITAL (H1)	*1.34*	*[1.28–1.4]*	*0.000***	*1.34*	*[1.28–1.40]*	*0.000***
GVC (H2)	*1.53*	*[1.30–1.81]*	*0.000***	*1.63*	*[1.36–1.95]*	*0.000***
FAIL				*1.39*	*[1.15–1.68]*	*0.001***
DIGITAL x FAIL (H3)				*0.98*	*[0.87–1.09]*	*0.688*
GVC × FAIL (H4)				*0.59*	*[0.37–0.96]*	*0.034***
Chi-squared test	969.43**			1,116.14**		
Pseudo R^2	5.8%			6.0%		
Hosmer–Lemeshow test	12798.12			12545.77		
Area under curve	67.8%			68.1%		

Note(s): OR – Odds Ratio; CI – Confidence Interval; *$p < 0.05$; **$p < 0.01$
Source(s): Authors' own creation

Table 4. Results for model 1 and model 2

(Model 1: $\chi^2 = 969.43$, $p < 0.01$; Model 2: $\chi^2 = 1{,}116.14$, $p < 0.01$), with no statistically significant differences between the estimated values and the observed values (Model 1: Hosmer–Lemeshow test = 12798.12, p = 0.0.766; Model 2: Hosmer–Lemeshow test = 12545.77, $p = 0.716$).

As for the control variables that have an impact on the creation of new business models, older companies are significantly less likely to implement new business models (Model 1: OR = 0.88, $p < 0.05$; Model 2: OR = 0.89, $p < 0.05$). Similarly, the higher the turnover, the lower the likelihood of implementing new business models (Model 1: OR = 0.99, $p < 0.01$; Model 2: OR = 0.99, $p < 0.01$). Conversely, the greater the number of employees, the greater the propensity to implement new business models (Model 1: OR = 1.20, $p < 0.01$; Model 2: OR = 1.20, $p < 0.01$). Companies in the accommodation and food service activities (Model 1: OR = 1.41, $p < 0.01$; Model 2: OR = 1.46, $p < 0.01$) and human health and social work activities (Model 1: OR = 1.53, $p < 0.01$; Model 2: OR = 1.53, $p < 0.01$) sectors were more likely to

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implement new business models than the manufacturing industry (we didn't include the dummy variable for the manufacturing industry, which was left as a reference category).

Hypothesis H1 "The digital capabilities of MSMEs have an impact on the adoption of new business models" shows that the higher the level of development of digital capabilities, the greater the propensity for business model innovation ((Model 1: OR = 1.34, $p < 0.01$; Model 2: OR = 1.34, $p < 0.01$). These results confirm Hypothesis 1.

As for Hypothesis 2, "Inclusion in global value chains by MSMEs has an impact on the adoption of new business models," there is a statistically significant effect of inclusion in global value chains on business model innovation (Model 1: OR = 1.53, $p < 0.01$; Model 2: OR = 1.63, $p < 0.01$), in which companies that are involved in global chains are more likely to implement new business models. These results confirm Hypothesis 2.

In H3, which analyses the moderating effect of the entrepreneur's previous failure on the impact of digital capabilities on the creation of new business models, the results show that the entrepreneur's failure has no statistically significant impact (Model 2: OR = 0.92, $p = 0.688$) on the relationship between digital capabilities and the creation of new business models. These results do not confirm Hypothesis 3.

Regarding Hypothesis H4 "The entrepreneur's previous failure moderates the impact of inclusion in global value chains of MSMEs on the adoption of new business models," entrepreneurial failure significantly alters the impact of inclusion in global value chains on business model innovation (Model 2: OR = 0.59, $p < 0.05$). In MSMEs where there has been some entrepreneurial failure, the impact of inclusion in global value chains on business model innovation is lower than in MSMEs where there has been no entrepreneurial failure. These results confirm Hypothesis 4.

6. Discussion

Digital capabilities play a significant role in business model innovation and, consequently, in firm performance. This research highlights that the higher the level of development of digital capabilities, the greater the propensity for innovation in business models. This finding is consistent with the dynamic capabilities perspective, which emphasizes the importance of integrating and reconfiguring internal and external competencies to deal with ever-changing business environments (Dejardin *et al.*, 2022; Ferreira *et al.*, 2020; Teece *et al.*, 1997). The integration of digital technology has transformed traditional dynamic capabilities, resulting in digital capabilities that impact business innovation and the creation of business models (Arranz *et al.*, 2023; Khurana *et al.*, 2022). Business model innovation acts as an intermediary in the relationship between digital capabilities and business performance (Anwar *et al.*, 2022; Wang *et al.*, 2023). Therefore, firms with strong digital capabilities are more likely to develop innovative and effective business models (Bresciani *et al.*, 2021; Wang *et al.*, 2023).

This research addresses the importance of interorganizational relationships, particularly business networks like GVCs. The results indicate that the inclusion of SMEs in global value chains has a statistically significant impact on business model innovation. These networks play a critical role in the resource-based view of firms, where no single company possesses all the necessary resources and capabilities to achieve its objectives (Ford *et al.*, 2011; Richardson, 1972). Therefore, firms access resources and capabilities through partnerships in their business networks (Barney, 1991, 2001; O'Farrell and Hitchens, 1988; Penrose, 2009). GVCs allow multinational firms in advanced economies to establish relationships with suppliers and other partner companies in developing and emerging countries (Kano *et al.*, 2020; Siaw and Okorie, 2022). However, digital technological advancements and the rise of the sharing economy are reshaping the basis of value offerings within GVCs, shifting from products and services to platforms (Nambisan *et al.*, 2019). These platforms facilitate interactions within multi-actor networks, enabling partner companies of multinational

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enterprises to participate in other networks, becoming both partners and competitors (Nambisan *et al.*, 2019). This shift is driving multinational firms to develop operational efficiencies and strategic effectiveness (Siaw and Okorie, 2022). Companies must be flexible in cooperation with their partners while differentiating from them. Additionally, they need to optimize value creation and capture by leveraging the specialized skills of their partners (Kano *et al.*, 2020; Siaw and Sarpong, 2021). Small and medium-sized enterprises (SMEs) participating in Global value chains benefit from involvement in multiple networks to learn and develop their capabilities, maximizing their competence in global markets (Ivarsson and Alvstam, 2010; Schmitz and Knorringa, 2000). Consequently, these SMEs are more inclined to adopt new business models, particularly given the technological advancements of I4.0 and the encouragement to engage in multiple platforms (Ivarsson and Alvstam, 2010; Schmitz and Knorringa, 2000). Therefore, the research underscores the importance of interactions in business networks like Global value chains in shaping SMEs' business models and how active participation in these networks and platforms contributes to adaptation and innovation in these models.

In this study, the entrepreneur's prior failure does not act as a significant moderator in the relationship between digital capabilities and business model innovation. This suggests that, within the context of this study, digital capabilities play a consistent and independent role in the creation of new business models, regardless of the entrepreneur's past failure history. This finding can be understood in light of the literature that emphasizes how resilient entrepreneurs view past failure as an opportunity for learning and the development of resilience, rather than a personal failure (Lafuente *et al.*, 2019) and how they can capitalize on the accumulated knowledge from their past entrepreneurial experiences (Coelho and McClure, 2005; Corner *et al.*, 2017; Khelil, 2016).

The results of this research provide a better understanding of how the experience of entrepreneurial failure moderates the impact of MSMEs inclusion in GVCs on the adoption of new business models. Within the context of GVCs, entrepreneurial failure triggers the release of "residual resources" that flow through the network toward new ventures (Petrucci and Milanesi, 2022). This implies that failure is not the end of the road but rather the beginning of a new phase filled with opportunities and changes (Petrucci and Milanesi, 2022). Experiences of failure can unpredictably shape the entrepreneurial trajectory, resulting in different interactions and resource recombination (Petrucci and Milanesi, 2022; Dana *et al.*, 2021). However, MSMEs that have previously experienced entrepreneurial failure are less inclined to embrace new business models compared to those without such setbacks. This can be attributed to the fact that prior failures have made these companies more averse to repeating similar mistakes in the future (Deichmann and Ende, 2014). They may opt for a more cautious approach to business model innovation, drawing from the lessons learned from past failures. Our findings underscore the complexity of entrepreneurship and how experiences of failure can influence future entrepreneurial decisions. MSMEs that have encountered failure may exhibit resilience and the ability to identify new opportunities (Deichmann and Ende, 2014), but they may also display hesitancy toward the adoption of new business models (Eklund *et al.*, 2020; Mai *et al.*, 2021).

6.1 Theoretical contributions

The theoretical implications drawn from this study are manifold and contribute significantly to our understanding of the intricate dynamics between entrepreneurial failure experience, the integration of MSMEs in GVCs, and the innovation of business models. Firstly, this research enhances our theoretical understanding by emphasizing the transformative impact of digital capabilities on traditional dynamic capabilities. As elucidated by Arranz *et al.* (2023) and Khurana *et al.* (2022), the integration of digital technology has given rise to digital capabilities,

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thereby influencing business innovation and the evolution of business models. The study builds on this theoretical foundation, positioning digital capabilities as pivotal drivers of innovative and effective business models (Bresciani *et al.*, 2021; Wang *et al.*, 2023). Furthermore, the study reinforces the theoretical significance of business networks, particularly GVCs, within the resource-based view of the firm. It aligns with the perspectives of Ford *et al.* (2011) and Richardson (1972), highlighting that firms, in their pursuit of objectives, rely on partnerships within their business networks to access the diverse resources and capabilities necessary for success. The research extends this theoretical framework to the evolving landscape of GVCs, where digital technological advancements and the emergence of the sharing economy reshape the foundations of value offerings (Nambisan *et al.*, 2019). The study underscores the need for companies to be adaptable, strategically effective and flexible in collaboration with partners within GVCs. Moreover, the study enriches the theoretical understanding of the post-failure phase in business networks. It acknowledges that entrepreneurial failure triggers the release of residual resources, a concept advanced by Petrucci and Milanesi (2022). This theoretical perspective challenges the notion that failure marks the end of a venture, emphasizing that it can initiate a phase filled with opportunities and changes. The study aligns with the literature on resilient entrepreneurs who perceive failure as an opportunity for learning and resilience development (Lafuente *et al.*, 2019). It further suggests that experiences of failure can shape the entrepreneurial trajectory unpredictably, influencing interactions and resource recombination.

6.2 Practical contributions

From a practical standpoint, MSMEs can leverage these results to enhance their participation in GVCs by adopting a more proactive approach in business networks. This can strengthen their learning capacity and skill development, enhancing their competitiveness in global markets. MSMEs need to be flexible in cooperating with partners and optimize value capture through specialized partner skills, especially in advanced technology and digital platform contexts. Additionally, adapting to new business models, particularly in response to technological advancements, can become a vital strategy for MSMEs success. Equally important is the provision of support for entrepreneurs who have experienced prior failures, including mentoring programs and resources to help them overcome the fear of failure and learn from their experiences. Encouraging entrepreneurs to view failure as an opportunity for learning and growth can assist MSMEs in planning more effective innovation strategies, considering entrepreneurs' prior experiences.

7. Final considerations, limitations and future lines of research

The primary aim of this research is to assess the connections among the digital capabilities of MSMEs, their integration into global value chains and the implementation of novel business models. Additionally, the study seeks to investigate how previous entrepreneurial failures might influence or moderate these connections.

While this study extends our comprehension of the relationships between digital capabilities, GVCs, entrepreneurial experiences and business model innovation in MSMEs, limitations should be considered when interpreting the findings. The research relies on data from the Flash Eurobarometer 486 survey, which is a comprehensive dataset that explores the challenges faced by European MSMEs. While this dataset is valuable, the results and relationships identified may not be fully generalizable to MSMEs outside the European context. This study examines entrepreneurial failure as a single variable, not exploring the nuances of the types of failure, severity or specific impacts of these failures. The study uses cross-sectional data, which has the limitation of providing a snapshot of the relationships

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during a specific period. Lastly, this study focuses on digital capabilities, inclusion in global value chains, and entrepreneurial failure. Other variables that may influence business model innovation, such as market conditions, competition or external impacts, are not considered in this study. Further research may be needed to address these limitations and advance this work, such as replicating the study in other regions or countries, as results may vary due to contextual differences. Exploring the nuances of failure types, severity or specific impacts and how they alter subsequent business decisions should be investigated through qualitative approaches. Finally, the use of longitudinal data, experimental studies or qualitative research could provide a deeper understanding of how these relationships evolve.

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(The Appendix follows overleaf)

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Appendix

	N	%
AT – Austria	457	3.3
BA – Bosnia and Herzegovina	192	1.4
BE – Belgium	470	3.4
BG – Bulgaria	474	3.4
CY – Cyprus (Republic)	198	1.4
CZ – Czech Republic	475	3.4
DE – Germany	464	3.3
DK – Denmark	473	3.4
EE – Estonia	481	3.4
ES – Spain	479	3.4
FI – Finland	477	3.4
FR – France	485	3.5
GB – United Kingdom	466	3.3
GR – Greece	477	3.4
HR – Croatia	475	3.4
HU – Hungary	481	3.4
IE – Ireland	449	3.2
IS – Iceland	190	1.4
IT – Italy	485	3.5
LT – Lithuania	478	3.4
LU – Luxembourg	187	1.3
LV – Latvia	487	3.5
MK – Makedonia/FYROM	190	1.4
MT – Malta	190	1.4
NL – The Netherlands	461	3.3
NO – Norway	293	2.1
PL – Poland	469	3.4
PT – Portugal	478	3.4
RO – Romania	475	3.4
RS – Serbia	189	1.4
RS-KM – Kosovo	194	1.4
SE – Sweden	468	3.4
SI – Slovenia	477	3.4
SK – Slovakia	487	3.5
TR – Turkey	275	2.0

Source(s): Authors' own creation

Table A1. Countries and firms

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